Probabilities from Venn Diagrams

Exam code: 0607

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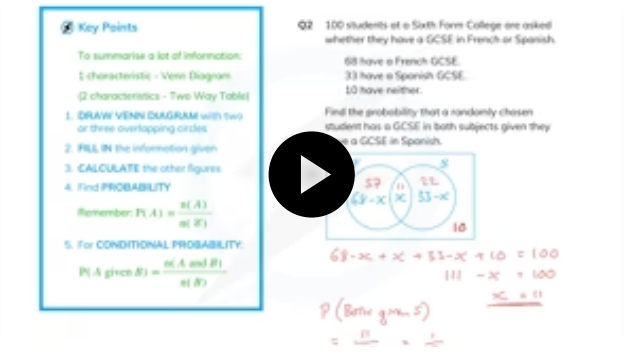

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Probability & Venn diagrams

How do I find probabilities from Venn diagrams?

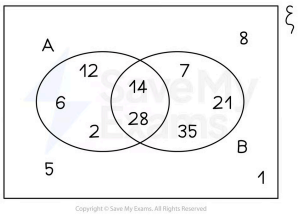
- If the Venn diagram shows **frequencies** in the region
  - **Add** together the frequencies in the regions you want
  - **Divide** by the total frequency
- If the Venn diagram shows the **individual elements**
  - **Count** the **number** of elements you want
  - **Divide** by the total number of elements
- For the Venn diagram shown below,
  - The probability of being **in *A*** is $\frac{5}{11}$
    - There are 5 elements in A out of 11 in total
  - The probability of being in **both** A **and** B is $\frac{2}{11}$
    - There are 2 elements in A and B (the intersection)
  - The probability of being **in *A***, but **not *B***, is $\frac{3}{11}$
    - 3 elements are in A but not B

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

already in A, is $\frac{2}{5}$

- You are only interested in elements in A
- There are 5 elements in A, out of which only 2 are also in B



Examiner Tips and Tricks

- Be careful when filling in numbers for a Venn diagram
  - Some of the given numbers may need to be split between two sections of the Venn diagram
- Suppose 10 people have a cat, 8 people have a dog and 6 people have both a cat and a dog
  - Out of the 10 people who have a cat
    - 6 also have a dog
    - 4 do not have a dog
  - Out of the 8 people who have a dog
    - 6 also have a cat
    - 2 do not have a cat

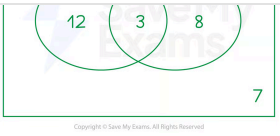
Worked Example

In a class of 30 students, 15 students study Spanish and 3 of the Spanish students also study German.
7 students study neither Spanish nor German.

(a) Draw a Venn diagram to show this information.

Answer:

Draw the Venn diagram with its rectangular box and two (labelled) overlapping circles
3 students study both Spanish and German, so start here and work outwards
12 must study Spanish but not German (to get 15 in total for Spanish)
7 study neither, so this goes outside of the circles
To get 30 in total, 8 must study German but not Spanish

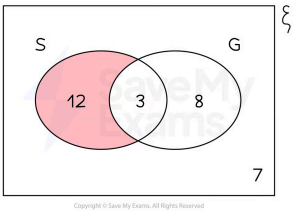


(b) Use your Venn diagram to find the probability that a student, selected at random from the class, studies Spanish but not German.

Answer:

This Venn diagram shows the frequencies in each region

It helps to highlight Spanish but not German



Divide the number of students studying Spanish but not German by the total number of students

Students studying Spanish but not German = 12
Total number of students = 30

$$\text{P(Spanish but not German)} = \frac{12}{30}\left(= \frac{2}{5}\right)$$

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